EXERCISE 4

# Symplectic Integrators for Orbits

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**1 Moon around the Earth.** Study the orbit of the Moon ($xy$ plane) under the gravity of the Earth (fixed at the origin):

$$\ddot{\mathbf{r}} = -\frac{GM}{|\mathbf{r}|^3}\,\mathbf{r}$$

with the mass of the Earth $M = 5.972 \cdot 10^{24}\,\mathrm{kg}$ and $G = 6.674 \cdot 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$.

**a)** Derive initial conditions $\mathbf{r}(0)$, $\mathbf{v}(0)$ for a circular orbit with radius $r_M = 384\,400\,\mathrm{km}$ and calculate its period $T$.
The sidereal month is 27.32 d. Where does the difference come from?

**b)** Implement different ODE methods (explicit Euler, implicit Euler, Heun, symplectic Euler and velocity Verlet, ...), simulate 10 orbits with $\Delta t = 6\,\mathrm{h}$ and plot them.
Which initial guess do you use for Newton, and how many iterations are needed?
Predict before running: which methods spiral outward, which inward?

**c)** Show that the energy $E = \frac{1}{2}|\mathbf{v}|^2 - GM/|\mathbf{r}|$ and the angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{v}$ (per unit mass) are conserved by the exact equation of motion. Plot the relative errors of $E$ and $L$ over 100 orbits ($\Delta t = 6\,\mathrm{h}$).
Which methods conserve $L$ exactly?
Why does Heun (order 2) end up worse than symplectic Euler (order 1)?

**d)** Calculate the error at $t = T/2$ (exact: $\mathbf{r} = (-r_M, 0)$) for $\Delta t = T/200, \ldots, T/6400$ and plot it log-log.
Which slopes do you get?
Which method would you use to simulate the Moon for 1000 years?

**2 Rocket launch.** A rocket starts at the equator of the Earth (radius $R_E = 6371\,\mathrm{km}$, rotation $\omega_E = 7.292 \cdot 10^{-5}\,\mathrm{s^{-1}}$). We simulate it in the equatorial plane (2D, Earth fixed at the origin, no atmosphere) with a constant thrust $F$ while the fuel burns ($0 \le t < t_b$, until the mass has dropped to $m_1$):

$$\ddot{\mathbf{r}} = -\frac{GM}{|\mathbf{r}|^3}\,\mathbf{r} + \frac{F}{m(t)}\,\mathbf{e}, \qquad m(t) = m_0 - \dot{m}\,t, \qquad F = \dot{m}\,v_e$$

and $F = 0$ afterwards, with $v_e = 4.4\,\mathrm{km/s}$. Start (co-rotating): $\mathbf{r} = (R_E, 0)$, $\mathbf{v} = (0, \omega_E R_E)$. Thrust direction $\mathbf{e} = \sin\theta\,\mathbf{e}_r \pm \cos\theta\,\mathbf{e}_\varphi$ ($+$ eastward, $-$ westward) with $\mathbf{e}_r = \mathbf{r}/|\mathbf{r}|$, $\mathbf{e}_\varphi = (-y, x)/|\mathbf{r}|$.

**a) Vertical launch.** Launch with $\theta = 90°$, $m_1 = m_0/19$ and $F/m_0 = 10, 12, 14, 16, 18\,\mathrm{m/s^2}$. Plot altitude and vertical speed over time, compare the speed at burnout with the rocket equation $\Delta v = v_e \ln(m_0/m_1)$ and check whether the rocket escapes ($E = \frac{1}{2}|\mathbf{v}|^2 - GM/|\mathbf{r}| \ge 0$ at burnout).
Where does the difference come from? Which thrust is needed to escape? Can a vertical launch reach an orbit?

**b) Pitch program.** Turn the rocket from vertical to horizontal: $\theta(t) = 90° \max(0, 1 - t/t_p)$, $F/m_0 = 15\,\mathrm{m/s^2}$, $m_1 = 0.12\,m_0$. Vary $t_p = 100, 125, \ldots, 300\,\mathrm{s}$, launch eastward and westward, simulate 3 h and classify the trajectories: crash, falls back, orbit or escape.
Which $t_p$ works best? Why do rockets launch eastward?

**c) Circular orbits.** With the rocket of b) the trajectories end on ellipses. How could a rocket like Falcon 9 reach circular orbits at $h = 200$, 500 and 1000 km? Develop a strategy (the engine may be switched off and relit), simulate it and plot the orbits.
How much fuel is left? Why does a single burn not work for all altitudes?

**d) Two stages.** Compare one stage (90 % fuel, $F/m_0 = 15\,\mathrm{m/s^2}$) with two stages (80 % + 20 % of $m_0$, each 90 % fuel). The empty first stage is dropped at burnout, the second one ignites with $F/m = 15\,\mathrm{m/s^2}$. Compare $\Delta v = \sum_k v_e \ln(m_{0,k}/m_{1,k})$ and plot the trajectories (eastward, best $t_p$ from b).
Why does staging help? How much $\Delta v$ is lost to gravity?